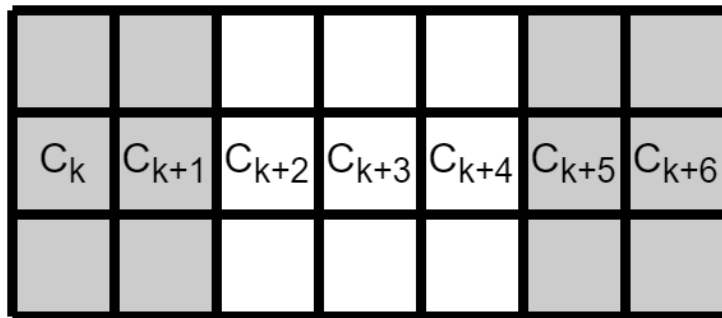


VLSI Design for Manufacturability HW4

Deadline: 2025/6/3

Instructions

- Ensure your handwriting is legible, or use a typed document if preferred.
 - Staple all pages together before submission.
1. Refer to p.1 to p.21, unit 10. The core mask covers the gray region in the figure below. Assume the minimum space for the core mask is 3. Answer the following questions.



- a. Give a Boolean expression for minimum core space rule. (10 points)
 - b. Convert the Boolean expression in (a) to CNF format. (10 points)
 - c. Convert the Boolean expression in (a) to ILP constraints. (10 points)
2. Refer to p.15, unit 11.
 - a. What is the n-wise method? (5 points) What are its key characteristics? (10 points)
 - b. Construct a 2-wise array with 5 columns where all values in the array are Boolean. (10 points)
 3. Refer to p.6 to p.21, unit 12.
 - a. Explain why $\frac{w_c + E}{2}$ is taken as the “effective width” of char c. (15 points)
 - b. In [1], they showed that the width of any tight packing is less than $E - 2S$ away from the minimum width packing. Explain why this bound holds. (20 points)
 4. Refer to p.52, unit 13, explain why Ours (DP) achieved shorter runtime compared to [7] DP-ILP. (10 points)

[1]: W.-K. Mak and C. Chu, "E-Beam Lithography Character and Stencil Co-Optimization," in IEEE TCAD, vol. 33, no. 5, pp. 741-751, May 2014, doi: 10.1109/TCAD.2013.2296496